

## Topic 6 Chapter 9 Worksheet

In the worksheets for this class, you are going to create MySQL statement for each task described below.

You are required to take a screen shot of the SQL statements and the output for each task.

Paste the screenshot of your results under the numbered question.

Save your work and upload the document to the assignment folder by the due date in the Syllabus.

### Database

Continue to use the Oracle tutorial database

1. Calculate the total for each order\_item (quantity \* unit\_price) and format the total to two digits

```
4 • SELECT order_id, FORMAT(quantity * unit_price, 2) AS total
5 FROM order_items;
```

| Result Grid |          |            | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|----------|------------|--------------|---------|--------------------|
|             | order_id | total      |              |         |                    |
| ▶           | 1        | 54,518.84  |              |         |                    |
|             | 1        | 52,898.23  |              |         |                    |
|             | 1        | 27,351.48  |              |         |                    |
|             | 1        | 89,609.13  |              |         |                    |
|             | 1        | 84,624.69  |              |         |                    |
|             | 1        | 60,894.05  |              |         |                    |
|             | 1        | 26,453.20  |              |         |                    |
|             | 1        | 49,533.42  |              |         |                    |
|             | 1        | 152,898.61 |              |         |                    |
|             | 1        | 77,320.53  |              |         |                    |
|             | 1        | 193,713.45 |              |         |                    |
|             | 1        | 31,914.55  |              |         |                    |
|             | 1        | 241,986.69 |              |         |                    |
|             | 2        | 33,186.78  |              |         |                    |
|             | 2        | 173,270.79 |              |         |                    |
|             | 2        | 41,183.61  |              |         |                    |
|             | 2        | 39,151.00  |              |         |                    |
|             | 2        | 57,261.90  |              |         |                    |
|             | 2        | 90,239.55  |              |         |                    |
|             | 2        | 68,699.00  |              |         |                    |
|             | 2        | 40,739.03  |              |         |                    |
|             | 2        | 46,571.25  |              |         |                    |
|             | 3        | 7,424.79   |              |         |                    |
|             | 3        | 1,104.35   |              |         |                    |

2. Using the employees table, concatenate the first to the last name with a comma, then the job\_title

```
8 • SELECT CONCAT_WS(' ', (CONCAT_WS(' ', first_name, last_name)), job_title) AS 'Employee/Title'
9 FROM employees;
```

| Result Grid    |                                             | Filter Rows: | Export: | Wrap Cell Content: |
|----------------|---------------------------------------------|--------------|---------|--------------------|
| Employee/Title |                                             |              |         |                    |
| ▶              | Tommy Bailey, President                     |              |         |                    |
|                | Jude Rivera, Administration Vice President  |              |         |                    |
|                | Blake Cooper, Administration Vice President |              |         |                    |
|                | Louie Richardson, Programmer                |              |         |                    |
|                | Nathan Cox, Programmer                      |              |         |                    |
|                | Gabriel Howard, Programmer                  |              |         |                    |
|                | Charles Ward, Programmer                    |              |         |                    |
|                | Bobby Torres, Programmer                    |              |         |                    |
|                | Mohammad Peterson, Finance Manager          |              |         |                    |
|                | Ryan Gray, Accountant                       |              |         |                    |
|                | Tyler Ramirez, Accountant                   |              |         |                    |
|                | Elliott James, Accountant                   |              |         |                    |
|                | Albert Watson, Accountant                   |              |         |                    |
|                | Elliot Brooks, Accountant                   |              |         |                    |
|                | Rory Kelly, Purchasing Manager              |              |         |                    |
|                | Alex Sanders, Purchasing Clerk              |              |         |                    |
|                | Frederick Price, Purchasing Clerk           |              |         |                    |
|                | Ollie Bennett, Purchasing Clerk             |              |         |                    |
|                | Louis Wood, Purchasing Clerk                |              |         |                    |
|                | Dexter Barnes, Purchasing Clerk             |              |         |                    |
|                | Liam Henderson, Stock Manager               |              |         |                    |
|                | Jackson Coleman, Stock Manager              |              |         |                    |

3. Using employees table, generate a login name according to these rules

Concatenate:

- first letter of the employee's first name
- to the last name
- to a random number that has been truncated to two digits
- and make the text upper case

```

12 • SELECT UPPER(CONCAT(
13         LEFT(first_name,1),
14         last_name,
15         LPAD(TRUNCATE(RAND()*100, 0), 2, '0')
16     )) AS Login
17 FROM employees;

```

Result Grid   Filter Rows:  | Export:  | Wrap Cell Content: 

|   | Login         |
|---|---------------|
| ▶ | TBAILEY65     |
|   | JRIVERA87     |
|   | BCOOPER40     |
|   | LRICHARDSON42 |
|   | NCOX89        |
|   | GHOWARD22     |
|   | CWARD41       |
|   | BTORRES42     |
|   | MPETERSON86   |
|   | RGRAY06       |
|   | TRAMIREZ70    |
|   | EJAMES33      |
|   | AWATSON57     |
|   | EBROOKS87     |
|   | RKELLY64      |
|   | ASANDERS61    |
|   | FPRICE10      |
|   | OBENNETT70    |
|   | LWOOD21       |
|   | DBARNES95     |

4. Using the customer table, what is the longest address?

```

20 • SELECT MAX(LENGTH(address)) AS 'the longest address'
21 FROM customers;

```

Result Grid   Filter Rows:  | Export:  | Wrap Cell Content: 

|   | the longest address |
|---|---------------------|
| ▶ | 47                  |

5. Using the customers table, locate the first comma in the address column whose name starts with a letter O

```
24 • SELECT name AS NAME, LOCATE(',', address)
25 FROM customers
26 WHERE REGEXP_LIKE(name, '^O');
```

| Result Grid          | Filter Rows:         | Export: | Wrap C |
|----------------------|----------------------|---------|--------|
| NAME                 | LOCATE(',', address) |         |        |
| Omnicom Group        | 18                   |         |        |
| Owens & Minor        | 16                   |         |        |
| Oracle               | 14                   |         |        |
| Office Depot         | 17                   |         |        |
| Occidental Petroleum | 22                   |         |        |

6. Using the products table, subtract standard cost from list price and round the total to zero decimal places

```
29 • SELECT product_name,
30 ROUND(list_price-standard_cost, 0) AS profit
31 FROM products;
```

| Result Grid              | Filter Rows: | Export: | Wrap Cell Content: |
|--------------------------|--------------|---------|--------------------|
| product_name             | profit       |         |                    |
| G.Skill Ripjaws V Series | 191          |         |                    |
| Intel Xeon E5-2697 V4    | 411          |         |                    |
| Corsair CB-9060011-WW    | 226          |         |                    |
| AMD 100-505989           | 571          |         |                    |
| PNY VCQK6000-PB          | 550          |         |                    |
| Zotac ZT-P10810A-10P     | 148          |         |                    |
| G.Skill Ripjaws V Series | 79           |         |                    |
| Intel Xeon E5-1650 V4    | 67           |         |                    |
| Intel Xeon E5-2640 V4    | 161          |         |                    |
| Crucial                  | 232          |         |                    |
| PNY VCQP5000-PB          | 413          |         |                    |
| Gigabyte GV-N108TAOR...  | 229          |         |                    |
| G.Skill Ripjaws V Series | 86           |         |                    |
| G.Skill Ripjaws V Series | 71           |         |                    |
| G.Skill Trident Z        | 124          |         |                    |
| Intel Core i7-6900K      | 97           |         |                    |

7. Take your birthdate and display the day of the week you were born

```
34 • SELECT DAYNAME(str_to_date('17.03.1981', '%d.%m.%Y')) AS 'My birth day of the week';
```

| Result Grid |                          | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|--------------------------|--------------|---------|--------------------|
|             | My birth day of the week |              |         |                    |
| ▶           | Tuesday                  |              |         |                    |

8. Use the CASE function to add an action item for each of the statuses in the order tables

If the status is shipped, then list the action item "send invoice"

If the status is pending, then the action item is "OK"

If the status is canceled, then the action item is "follow up"

```
37 • SELECT order_id, customer_id, status,  
38 CASE status  
39 WHEN 'Shipped' THEN 'send invoice'  
40 When 'Pending' THEN 'OK'  
41 WHEN 'Canceled' THEN 'follow up'  
42 END AS 'action needed'
```

| Result Grid |             | Filter Rows: | Export:       | Wrap Cell Content: |
|-------------|-------------|--------------|---------------|--------------------|
| order_id    | customer_id | status       | action needed |                    |
| ▶ 1         | 4           | Pending      | OK            |                    |
| 2           | 4           | Shipped      | send invoice  |                    |
| 3           | 5           | Shipped      | send invoice  |                    |
| 4           | 8           | Shipped      | send invoice  |                    |
| 5           | 5           | Canceled     | follow up     |                    |
| 6           | 6           | Shipped      | send invoice  |                    |
| 7           | 7           | Shipped      | send invoice  |                    |
| 8           | 8           | Shipped      | send invoice  |                    |
| 9           | 9           | Shipped      | send invoice  |                    |
| 10          | 44          | Pending      | OK            |                    |
| 11          | 45          | Shipped      | send invoice  |                    |
| 12          | 46          | Shipped      | send invoice  |                    |
| 13          | 47          | Shipped      | send invoice  |                    |
| 14          | 48          | Shipped      | send invoice  |                    |
| 15          | 49          | Shipped      | send invoice  |                    |
| 16          | 16          | Pending      | OK            |                    |
| 17          | 17          | Shipped      | send invoice  |                    |
| 18          | 18          | Shipped      | send invoice  |                    |

9. Use the IF statement to add a column to the query results showing whether a country is US or not

Show all locations

```

47 • SELECT location_id, country_id,
48         IF(country_id='US', 'Domestic', 'International') AS Travel
49 FROM locations;

```

Result Grid   Filter Rows:  Export:  Wrap Cell Content: 

|   | location_id | country_id | Travel        |
|---|-------------|------------|---------------|
| ▶ | 13          | AU         | International |
|   | 19          | BR         | International |
|   | 9           | CA         | International |
|   | 10          | CA         | International |
|   | 20          | CH         | International |
|   | 21          | CH         | International |
|   | 11          | CN         | International |
|   | 18          | DE         | International |
|   | 12          | IN         | International |
|   | 1           | IT         | International |
|   | 2           | IT         | International |
|   | 3           | JP         | International |
|   | 4           | JP         | International |
|   | 23          | MX         | International |
|   | 22          | NL         | International |
|   | 14          | SG         | International |
|   | 15          | UK         | International |
|   | 16          | UK         | International |
|   | 17          | UK         | International |
|   | 5           | US         | Domestic      |
|   | 6           | US         | Domestic      |
|   | 7           | US         | Domestic      |
|   | 8           | US         | Domestic      |

10. Using the employees table, calculate the number of years from hire date to now

```
52 • SELECT hire_date,  
53         CURRENT_DATE AS 'current_date',  
54         Year(CURRENT_DATE()) - YEAR(hire_date) AS 'length of employment'  
55 FROM employees;
```

Result Grid   Filter Rows:  | Export:  | Wrap Cell Content: 

|   | hire_date  | current_date | length of employment |
|---|------------|--------------|----------------------|
| ▶ | 2016-06-17 | 2026-10-03   | 10                   |
|   | 2016-09-21 | 2026-10-03   | 10                   |
|   | 2016-01-13 | 2026-10-03   | 10                   |
|   | 2016-01-03 | 2026-10-03   | 10                   |
|   | 2016-05-21 | 2026-10-03   | 10                   |
|   | 2016-06-25 | 2026-10-03   | 10                   |
|   | 2016-02-05 | 2026-10-03   | 10                   |
|   | 2016-02-07 | 2026-10-03   | 10                   |
|   | 2016-08-17 | 2026-10-03   | 10                   |
|   | 2016-08-16 | 2026-10-03   | 10                   |
|   | 2016-09-28 | 2026-10-03   | 10                   |
|   | 2016-09-30 | 2026-10-03   | 10                   |
|   | 2016-03-07 | 2026-10-03   | 10                   |
|   | 2016-12-07 | 2026-10-03   | 10                   |
|   | 2016-12-07 | 2026-10-03   | 10                   |
|   | 2016-05-18 | 2026-10-03   | 10                   |
|   | 2016-12-24 | 2026-10-03   | 10                   |
|   | 2016-07-24 | 2026-10-03   | 10                   |
|   | 2016-11-15 | 2026-10-03   | 10                   |
|   | 2016-08-10 | 2026-10-03   | 10                   |
|   | 2016-04-10 | 2026-10-03   | 10                   |
|   | 2016-05-01 | 2026-10-03   | 10                   |

11. Create a query that displays the rank and dense rank for the credit limit of customers  
Use the customer table

Order by credit limit

319 rows returned

```

58 • SELECT RANK() OVER (ORDER BY credit_limit DESC) AS 'rank',
59         DENSE_RANK() OVER(ORDER BY credit_limit DESC) AS 'dense rank',
60         credit_limit, customer_id, name
61 FROM customers;

```

| Result Grid   Filter Rows: <input type="text"/>   Export:    Wrap Cell Content:  |      |            |              |             |                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------|--------------|-------------|-------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                        | rank | dense rank | credit_limit | customer_id | name                                |
| ▶                                                                                                                                                                                                                                                                                                                                                                                                                      | 1    | 1          | 5000.00      | 173         | Massachusetts Mutual Life Insurance |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 174         | Orade                               |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 175         | Morgan Stanley                      |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 176         | Cigna                               |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 177         | United Continental Holdings         |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 178         | Allstate                            |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 179         | TIAA                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 180         | INTL FCStone                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 181         | CHS                                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 182         | American Express                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 183         | Gilead Sciences                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 186         | TJX                                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 219         | Duke Energy                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 222         | GameStop                            |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 223         | Northrop Grumman                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 224         | Arrow Electronics                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 226         | Office Depot                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 227         | Nordstrom                           |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 228         | Kinder Morgan                       |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 234         | Visa                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 235         | Lincoln National                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 236         | Ecolab                              |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 240         | Loews                               |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 289         | BlackRock                           |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 290         | WestRock                            |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 291         | Voya Financial                      |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 301         | Univar                              |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 302         | D.R. Horton                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 303         | Estee Lauder                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                        | 1    | 1          | 5000.00      | 309         | Reynolds American                   |

A little further down the list

```

58 • SELECT RANK() OVER (ORDER BY credit_limit DESC) AS 'rank',
59         DENSE_RANK() OVER(ORDER BY credit_limit DESC) AS 'dense rank',
60         credit_limit, customer_id, name
61 FROM customers;

```

| Result Grid     Filter Rows: <input type="text"/>   Export:    Wrap Cell Content:  |      |            |              |             |                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------|--------------|-------------|--------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                          | rank | dense rank | credit_limit | customer_id | name                           |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 1    | 1          | 5000.00      | 310         | Core-Mark Holding              |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 1    | 1          | 5000.00      | 312         | Henry Schein                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 1    | 1          | 5000.00      | 313         | Hertz Global Holdings          |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 34   | 2          | 3700.00      | 166         | Best Buy                       |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 34   | 2          | 3700.00      | 167         | Merck                          |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 34   | 2          | 3700.00      | 168         | Advance Auto Parts             |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 34   | 2          | 3700.00      | 169         | Liberty Mutual Insurance Group |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 34   | 2          | 3700.00      | 170         | Goldman Sachs Group            |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 34   | 2          | 3700.00      | 171         | Honeywell International        |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 34   | 2          | 3700.00      | 172         | MasterCard                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 34   | 2          | 3700.00      | 210         | CST Brands                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 34   | 2          | 3700.00      | 211         | McDonald's                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 34   | 2          | 3700.00      | 220         | Time Warner Cable              |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 34   | 2          | 3700.00      | 225         | Reliance Steel & Aluminum      |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 34   | 2          | 3700.00      | 244         | Viacom                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 34   | 2          | 3700.00      | 248         | Devon Energy                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 34   | 2          | 3700.00      | 249         | PBF Energy                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 48   | 3          | 3600.00      | 157         | Ingram Micro                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 48   | 3          | 3600.00      | 158         | Owens & Minor                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 48   | 3          | 3600.00      | 159         | Energy Transfer Equity         |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | 48   | 3          | 3600.00      | 160         | Tyson Foods                    |